Qualitative Coding of Scientific Communities in the Literature Selection

Inductive thematic categorization based on the CSV columns **title** and **keywords**. The purpose is to identify which scientific communities are represented in the literature selection.

The dataset contains 250 records. Titles and keywords were combined into one text field per record, normalized, and reviewed using an inductive qualitative coding logic supported by keyword-frequency inspection and clustering of title-keyword language. Categories are thematic and non-exclusive: one publication can indicate several communities, for example higher education plus AI or STEM plus assessment. Counts therefore represent indicative mentions, not mutually exclusive publication totals. The most frequent raw keyword terms included: higher education (19), covid-19 (10), stem education (10), motivation (8), assessment (7), virtual reality (7), inclusive education (7), artificial intelligence (6), education (6), learning analytics (6), systematic review (6), children (5), stress (4), intervention (4), science communication (4), self-efficacy (4), psychometrics (4), undergraduate education (4), learning (4), teacher education (4).

Category overview

No.	Community category	Definition	Indicative records	Example from the dataset
1	Educational Technology, AI, Learning Analytics and Online Assessment	Research communities examining digital platforms, artificial intelligence, generative AI, learning analytics, online learning environments, proctoring, educational data science, and technology-enhanced assessment.	58	**ChatGPT: The End of Online Exam Integrity?** *Keywords:* AI impact on education; ChatGPT in exam cheating; multimodal online assessments; GPT-4 vision evaluation; educational integrity; generative AI *Year:* 2024
2	STEM, Science, Mathematics and Computing Education	Research communities focused on STEM learning and teaching, including mathematics, science, biology, computer science, programming, engineering, physics-related topics, and STEM identity or outreach.	42	**Testing the impact of two afterschool museum outreach interventions on elementary childrens STEM outcomes: hands-on STEM alone or with STEM stories** *Keywords:* informal learning; science, technology, engineering and math (STEM); museum education; family involvement; Expectancy-Value Theory; achievement *Year:* 2025
3	Higher Education, Graduate Education and Academic Careers	Communities studying universities, college students, doctoral training, graduate curricula, campus experiences, career outcomes, student evaluations, and academic transitions into or through higher education.	52	**Going to College With a Posse: How Having High School Peers on Campus Supports College Achievement** *Keywords:* admissions; equity; first-generation college students; hierarchical linear modeling; higher education; peer interaction/friendship; privilege; psychology; regression analyses; retention; social belonging; social capital; socioeconomic status *Year:* 2021
4	Teacher Education, Professional Development and Teacher Communities	Research on teacher preparation, in-service professional development, professional identity, teacher knowledge, teacher attitudes, teacher learning networks, and research-practice communication with educators.	46	**Rethinking Professional Development: Development and Evaluation of an Evidence-Based Online PD Course on the Effective Use of Technology in the Classroom** *Keywords:* online professional development; ICAP; in-service teacher training; prepost design; teachers technological pedagogical knowledge *Year:* 2025
5	Inclusion, Special Education, Equity and Diversity	Communities addressing inclusive education, special educational needs, disability, accessibility, equity, first-generation or underrepresented students, socioeconomic status, gender, race, ethnicity, and diversity in educational contexts.	37	**Going to College With a Posse: How Having High School Peers on Campus Supports College Achievement** *Keywords:* admissions; equity; first-generation college students; hierarchical linear modeling; higher education; peer interaction/friendship; privilege; psychology; regression analyses; retention; social belonging; social capital; socioeconomic status *Year:* 2021

No.	Community category	Definition	Indicative records	Example from the dataset
6	Literacy, Reading, Language and Bilingualism	Research communities concerned with reading fluency, comprehension, vocabulary, language learning, bilingualism, dyslexia, foreign-language learning, writing, and text-based literacy development.	38	**Development and validation of a rapid and precise online sentence reading efficiency assessment** *Keywords:* dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics *Year:* 2024
7	Psychology of Learning, Motivation, Well-Being and Socioemotional Development	Communities using psychological constructs to study learning, including motivation, self-efficacy, self-regulation, emotions, stress, well-being, mental health, personality, belonging, engagement, and socioemotional skills.	58	**Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources** *Keywords:* COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories *Year:* 2022
8	Assessment, Measurement, Psychometrics and Evaluation	Communities developing or evaluating instruments, scales, tests, diagnostics, assessment systems, psychometric models, program evaluations, and measurement approaches.	76	**A New Measurement Instrument for Music-Related Argumentative Competence: The MARKO Competency Test and Competency Model** *Keywords:* music-related argumentation; competency; assessment; music; reasoning; item response theory; empirical research; musical judgment *Year:* 2021
9	Curriculum, Textbooks and Educational Content Studies	Research communities analyzing curricula, textbooks, instructional materials, lesson study, content representation, educational resources, and subject-matter organization.	10	**Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks** *Keywords:* artificial intelligence; case studies; content analysis; curriculum; data science; gender studies; history; natural language processing; race; textbooks; textual analysis *Year:* 2020
10	Open Science, Research Transparency, OER and Science Communication	Communities concerned with openness and credibility in education research, including preregistration, open science, open educational resources, registered reports, replication, evidence communication, and science communication training.	18	**Open Education Science** *Keywords:* preregistration; open science; registered report; open access *Year:* 2018
11	COVID-19, Distance Learning and Home Education	Communities studying educational disruptions and adaptations during the COVID-19 pandemic, including lockdown learning, distance education, home education, online transitions, stress, and family or parental perspectives.	18	**Mandatory Home Education During the COVID-19 Lockdown in the Czech Republic: A Rapid Survey of 1st-9th Graders Parents** *Keywords:* COVID-19; distance education; home education; online education; parents; Czech Republic *Year:* 2020
12	Immersive, Game-Based and Experiential Learning Environments	Communities researching virtual reality, augmented reality, metaverse environments, serious games, gamification, simulations, museum outreach, and other experiential or interactive learning designs.	18	**Integrating immersive technologies with STEM education: a systematic review** *Keywords:* immersive technologies; virtual reality; augmented reality; STEM education; PICOS approach; PRISMA guidelines; teaching and learning *Year:* 2024

Interpretive synthesis

The literature selection is strongly shaped by education research communities that connect empirical educational studies with technology-oriented methods and open-science practices.

The largest visible communities are educational technology/AI/learning analytics, higher education, teacher education, STEM/science education, assessment/measurement, and psychology of learning. These communities overlap substantially, especially where AI or digital tools are used for assessment, teacher professional learning, STEM learning, or higher education governance.

A second group of communities is defined by learner populations and social concerns: inclusion and special education, equity and diversity, reading/literacy, and well-being. These categories suggest that the corpus is not only methodologically or technologically oriented but also concerned with access, participation, and learner support.

Open science, OER, research transparency, and science communication form a distinct cross-cutting community. It is smaller than the technology or higher-education clusters, but it is strategically important for the overall aim of identifying open data, open code, open materials, preregistration, and transparent research practices.

Coding notes and limitations

- The categories are content-based and derived only from title and keyword information, not from full-text reading.

- Because the coding is non-exclusive, counts should be interpreted as evidence of community presence rather than as a distribution of publications across mutually exclusive fields.

- 22 records were not captured by the rule-based category labels after qualitative consolidation. This does not mean they are irrelevant; rather, their title-keyword wording was less aligned with the final community labels.

- Examples are representative records selected from the dataset to illustrate each category. They are not necessarily the most important or most cited publication in that community.

Suggested use of the category system

For subsequent coding, these categories can be used as a non-exclusive coding frame. A paper may receive multiple community codes when its title and keywords show clear overlap, such as *AI in higher education assessment*, *inclusive STEM education*, or *teacher professional development and science communication*.

Indicative category counts, sorted by frequency

Category	Indicative records
Assessment, Measurement, Psychometrics and Evaluation	76
Educational Technology, AI, Learning Analytics and Online Assessment	58
Psychology of Learning, Motivation, Well-Being and Socioemotional Development	58
Higher Education, Graduate Education and Academic Careers	52

Category	Indicative records
Teacher Education, Professional Development and Teacher Communities	46
STEM, Science, Mathematics and Computing Education	42
Literacy, Reading, Language and Bilingualism	38
Inclusion, Special Education, Equity and Diversity	37
Open Science, Research Transparency, OER and Science Communication	18
COVID-19, Distance Learning and Home Education	18
Immersive, Game-Based and Experiential Learning Environments	18
Curriculum, Textbooks and Educational Content Studies	10